

Multiple Choice (10 marks)

1. For $f(t) = e^{-2t}u(t)$, find the initial value and slope both by letting $t = 0^+$ and from the initial-value theorem.
2. Initial value is 2, slope is -4
3. Initial value is -1, slope is 2
4. Initial value is 1, slope is 2
5. Initial value is 0, slope is undefined
6. Initial value is 0, slope is 2

Find the input impedance of a 50-ohm line terminated in $+j 50$ ohms, for a line length such that $\beta l = (\pi/1)$ radian.

- (a) 100 ohms
- (b) 25 ohms
- (c) $j 100$ ohms
- (d) 50 ohms
- (e) $-j 50$ ohms

The magnetisation and applied field in ferromagnetic materials are related

- (a) linearly.
- (b) cubically.
- (c) parabolically.
- (d) logarithmically.
- (e) hyperbolically.

A Binary number system has how many digits.

- (a) 0
- (b) 1
- (c) 2
- (d) 10

What is Newton's Law of cooling as applied to convective heat transfer?

- (a) $q = hA$
- (b) $q = h^2 A$
- (c) $q = hA(T - T_0)$
- (d) $q = h / A$
- (e) $q = hA/T$

True / False (10 marks)

Answer True or False

6. True or False: The impact stress in the steel rod under the given conditions is 25,678.9 psi.
7. True or False: The determinant of matrix A is 2, suggesting that the matrix is invertible and has a non-zero determinant.
8. True or False: The solution to the difference equation is $y[n] = (27/20) 3^n - (3/5) (1/2)^n + (1/4) (1/3)^n$.
9. True or False: The probability $\Pr[0 < x < (3/4), (1/3) < y < 2]$ is equal to $3/4$ for the given joint density function $f(x, y)$.
10. True or False: The Laplace transform of $f(t) = t^2$ is given by $L\{f(t)\} = 2/s^3$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Derive the homogeneous solution for the voltage $v_C(t)$ across a $(1/2)$ F capacitor in a series RLC circuit with $R = 4$, $L = 4$ H, initial conditions $i_L(0) = 1/4$ A and $v_C(0) = 0$.
12. Discuss the impact of varying the modulation index (β) on the signal power passed by the filter in a space communication system, specifically analyzing the cases where $\beta = 0.2$ and $\beta = 5$.

End of Paper